

Data Product Manager Nanodegree

Applying Data Science to Product Management

Final Project: Developing an MVP Launch Strategy for a Flying Taxi Service

Welcome to your first week at Flyber

Flyber

In this project, you will apply the skills acquired in this course to create the MVP launch strategy for the first flying car taxi service, Flyber, in one of the most congested cities in America -- New York City.

You are responsible for bringing the first flying car taxi service to market by analyzing data and building a product proposal.

You will need to use the SQL workspace provided in the Classroom, and [Tableau Public](#), in order to successfully complete the project.

You'll present your answers, findings, and insights in the Answer Slides found in this deck. Feel free to include any additional slides, if needed.

Section 1: Data Exploration

Back to the basics of product management, identify your customer and their pain points:

- What are taxis used for?
- What are the characteristics of the users that leverage them?
- What are existing pain points with taxis?
- What are the existing pain points with digital ride-sharing services?

Ground travel creates uncertainty

USES: commuting, airport transfers, appointments, nightlife, tourism, and transit-gap trips.

USERS: residents and visitors without cars, time-sensitive riders, and people needing door-to-door access.

TAXI PAINS: road congestion, uncertain arrival time, variable availability, fare opacity, and driver idle/deadhead time.

RIDESHARE PAINS: surge pricing, cancellations, pickup mismatch, driver pay volatility, curb congestion, and privacy concerns.

What user improvements do you hypothesize a flying taxi service would have over the existing state of taxis today?

What market improvements do you hypothesize a flying taxi service would have the existing taxi service industry & physical road infrastructure today?

Flyber's advantage is time certainty

USER HYPOTHESIS: bypassing road bottlenecks will reduce travel-time variance on short, slow trips and make airport/business-district transfers more predictable.

DRIVER/PILOT HYPOTHESIS: scheduled nodes and reservations can reduce curbside searching, cancellations, and unpaid deadhead time.

MARKET HYPOTHESIS: shifting selected trips off roads may reduce localized curbside and road demand, but could create noise, airspace, safety, and access externalities.

These are testable hypotheses, not promised outcomes.

Upload [this dataset](#) into Tableau Online.

Ensure the fields are parsed correctly; field headers are included in the first row of the CSV.

Let's begin exploration!

Acquire a high-level understanding of the granularity and scope of the dataset, to inform the basis for your analyses:

- How many records are in the dataset
- What does each record represent?
- What is the primary key?
- What date range is your dataset bound to?
- What are the geographical bounds of this dataset? Is it limited to Manhattan, or is Brooklyn, Queens, Staten Island, the Bronx, and New Jersey included? Where are most of the data points centralized at? Are there outliers?

1,048,468 unique trips define the market

GRAIN: one completed taxi trip with pickup/dropoff time and coordinates, passengers, duration, and distance.

PRIMARY KEY: id; 1,048,468 records and 1,048,468 unique ids.

TIME: pickups from Jan 1 through Jun 30, 2016; the latest dropoff is Jul 1, 2016.

SPACE: trips center on Manhattan and extend to outer borough/airport catchments. Raw outlier bounds reach 32.18-51.88 N and 79.82-61.34 W.

QUALITY: 1,031,441 records (98.4%) pass the documented operational filter; 99.9% of pickups and 99.6% of dropoffs map to an NYC NTA.

You notice that the dataset does not contain explicit data points out-of-the-box, we'll need to enrich the dataset with relevant fields:

- You notice that ride price is not included, but figure it could be derived. Based on information about New York taxi prices gleaned from the internet, create a calculated field called `price` using the `duration`, `distance`, and `passenger count` fields.
- You hypothesize your target users will be those who take a relatively longer time getting to a destination that is relatively close, due to heavy traffic conditions and/or limitations to physical road infrastructure. To be able to analyze where this is happening, you will need to create a calculated field called `duration-to-distance ratio`.

Let's understand the scope and distribution various dimensions within the dataset. Calculate the **average**, **median**, and the **first & second standard deviation of the mean** for the following measures:

- duration
- distance
- passenger counts
- duration-to-distance ratio
- price

A typical trip is 11 min over 2.1 mi

Operational-quality subset: $n = 1,031,441$ (98.4% of records)

Duration (min): mean 14.04 | median 11.12 | SD 10.85

1 SD: 3.19 to 24.89 | 2 SD: -7.67 to 35.71

Distance (mi): mean 3.48 | median 2.13 | SD 3.92

1 SD: -0.44 to 7.41 | 2 SD: -4.37 to 11.33

Passengers: mean 1.66 | median 1.00 | SD 1.31

1 SD: 0.35 to 2.98 | 2 SD: -0.96 to 4.29

Minutes/mile: mean 5.39 | median 4.66 | SD 3.33

1 SD: 2.06 to 8.71 | 2 SD: -1.26 to 12.04

Fare proxy (USD): mean 12.94 | median 9.83 | SD 9.70

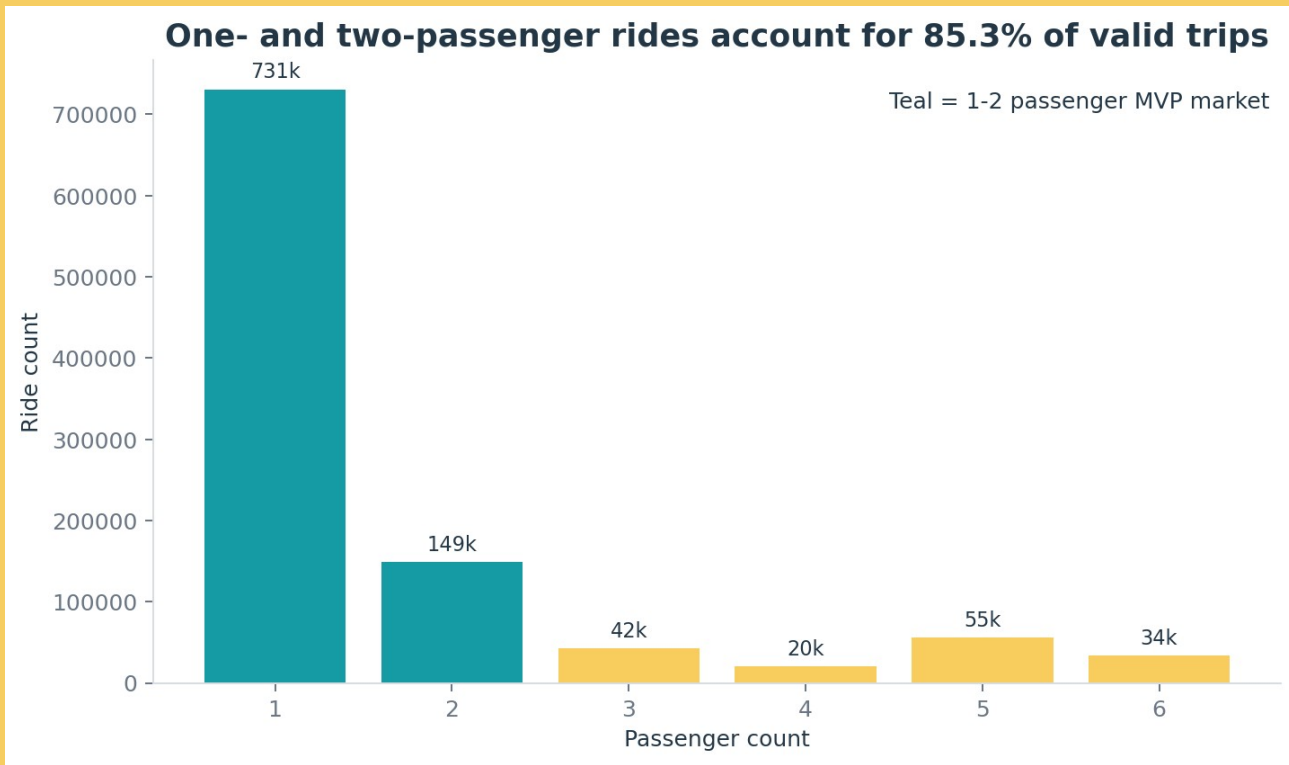
1 SD: 3.24 to 22.65 | 2 SD: -6.46 to 32.35

SD bands are descriptive and may cross impossible values; medians are retained for skewed measures.

Flying cars may have to have to be a lower weight for efficiency & take-off. Or you may just decide to leverage mini-copters for your initial MVP.

Create a histogram that visualizes the number of total rides grouped by passenger counts to analyze the potential market volume of low passenger pickups (1-2 passengers).

1-2 riders represent 85.3% of trips

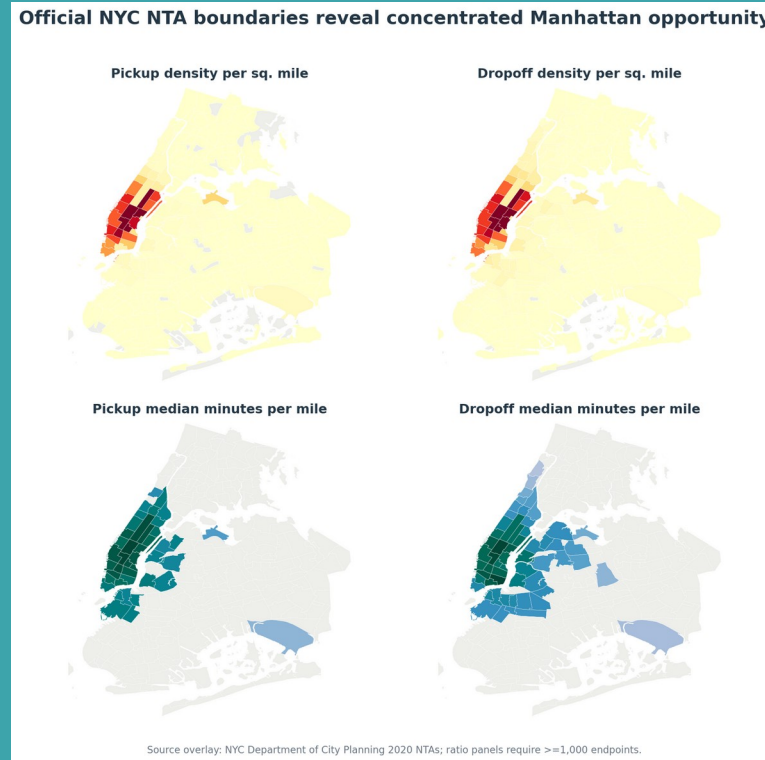


Full-data analysis; operational-quality subset $n = 1,031,441$.

For the initial MVP launch (& most likely GA), we have a finite amount of monetary resources to build Flyber pick-up / drop-off nodes. We'll need to be strategic on where we'll place them:

- Which neighborhoods/zip codes tends to experience a relatively higher density of pick-ups?
- Which neighborhoods/zip codes tends to experience a relatively higher density of drop-offs?
- Which neighborhoods/zip codes tends to have the highest duration-to-distance ratios, based on pick-up?
- Which neighborhoods/zip codes tends to have the highest duration-to-distance ratios, based on drop-off?
- For any of the neighborhoods identified, are there any potential areas within the neighborhood that are optimal for flying taxi pick-up / drop-off? What makes them suitable?

Midtown concentrates demand and delay



E34th/W30th: Midtown; Pier 6: Downtown; LGA: airport. Existing sites limit build cost. [Open Tableau Public workbook.](#)

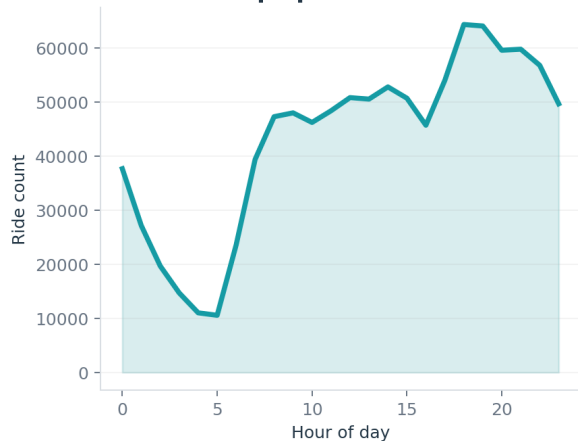
It may not make operational sense to have the service running 24/7, for now.

- What times throughout the day experience relatively higher volumes of ride pick-ups?
- What days throughout the week experience relatively higher volumes of ride pick-ups?
- Pinpoint any periods throughout the year that experience trend fluctuation or seasonality around ride pick-up volumes. This will help us in our post-launch analyses to determine if any spikes or dips were influenced by seasonality or through actual feature adoption/regression.

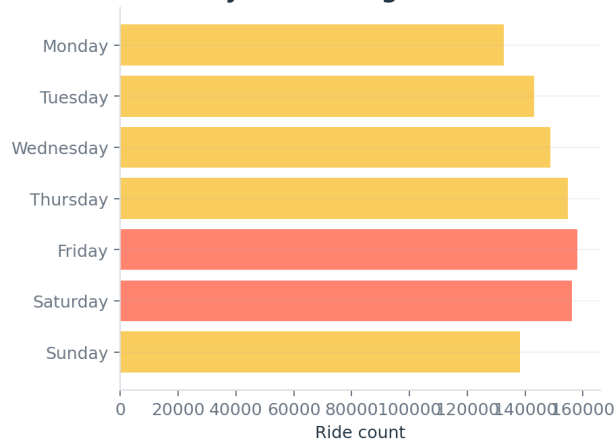
Thu-Sat evenings maximize demand

Temporal demand supports a Thursday-Saturday evening MVP

Pickups peak at 6-7 PM



Friday has the highest volume



H1 volume peaks in March



H1 peaks in March; six months cannot establish annual seasonality.

You and the user research team ran a quantitative survey on existing taxi and/or rideshare users in New York City to determine sentiment around potentially using a flying taxi service.

Dive into the survey results dataset in order to extract insights from explicit feedback.

Upload [this dataset](#) into Tableau Online or a SQL database (the classroom contains a workspace with the data for you as well).

Ensure the fields are parsed correctly, field headers are included in the first row of the CSV.

Question schema:

Q1 - What is your email?

Q2 - What gender do you identify as?

Q3 - What is your age?

Q4 - What is your annual income? (income bands)

Q5 - What neighborhood do you reside in?

Q6 - Do you currently use taxis? (Y/N)

Q7 - Do you currently use ridesharing services? (Y/N)

Q8 - Would you use a flying taxi service, if such a concept existed? (Y/N)

Q9 - If yes to Q8, how much would you be willing to pay per mile for such a service? (USD)

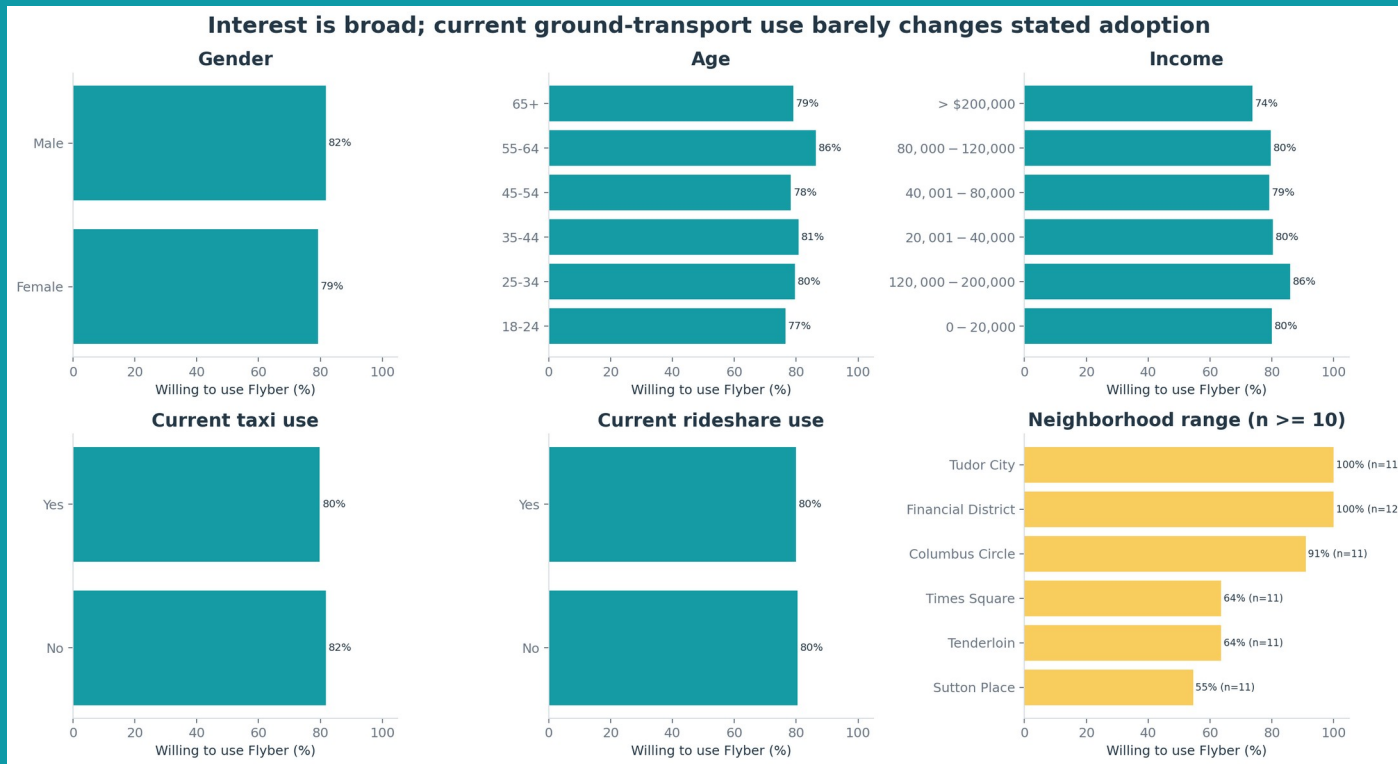
Q10 - If no to Q8, what is the reason?

To inform our future product marketing efforts, we'll want to extract the following:

- Is there an inclination of better Flyber adoption based on gender, age, income level, or neighborhood of residence?
- What is the distribution of potential price per mile based on gender, age, income level, and neighborhood of residence?
- What is the different personas/segments of negative sentiment towards not using a flying taxi car service?

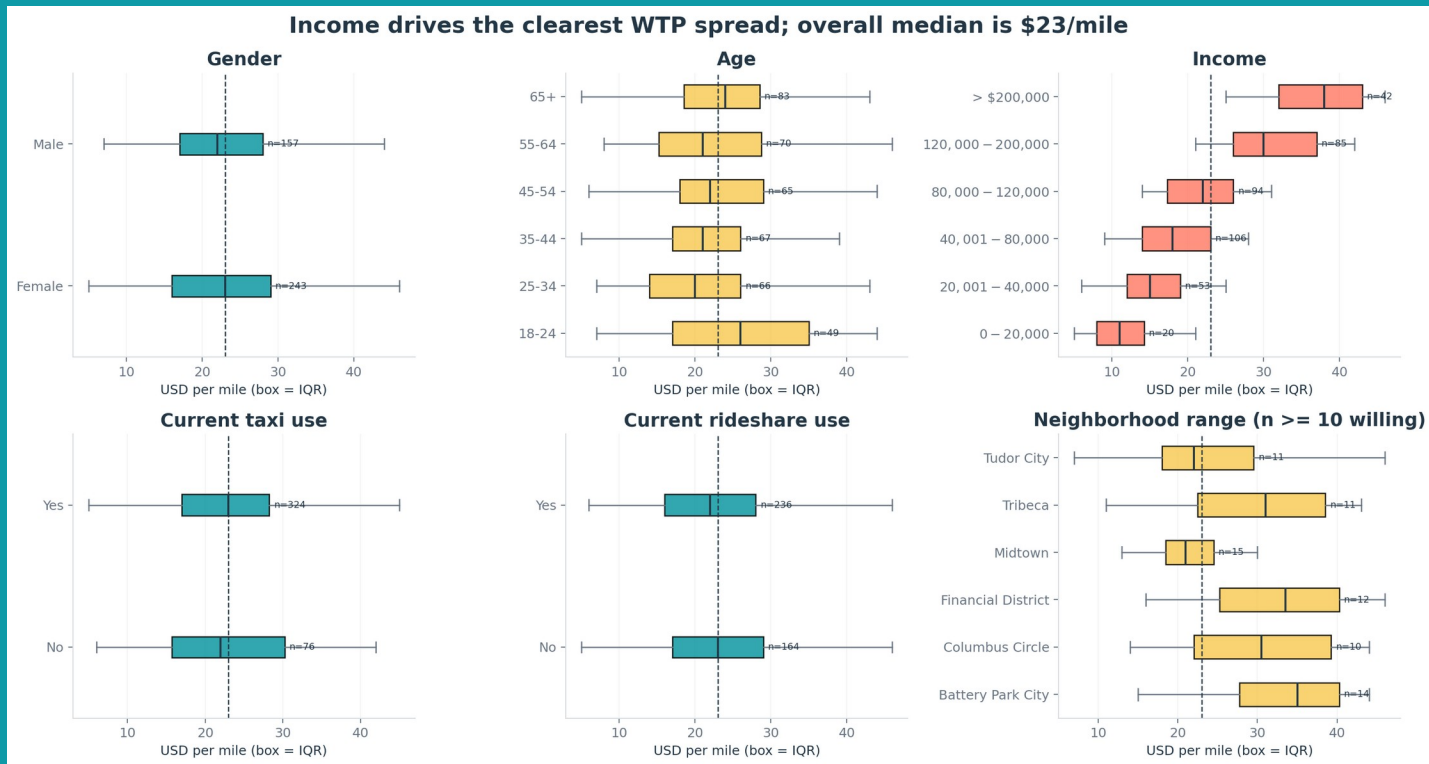
Please support your answers with clear, relevant visualizations.

Transport use barely changes interest



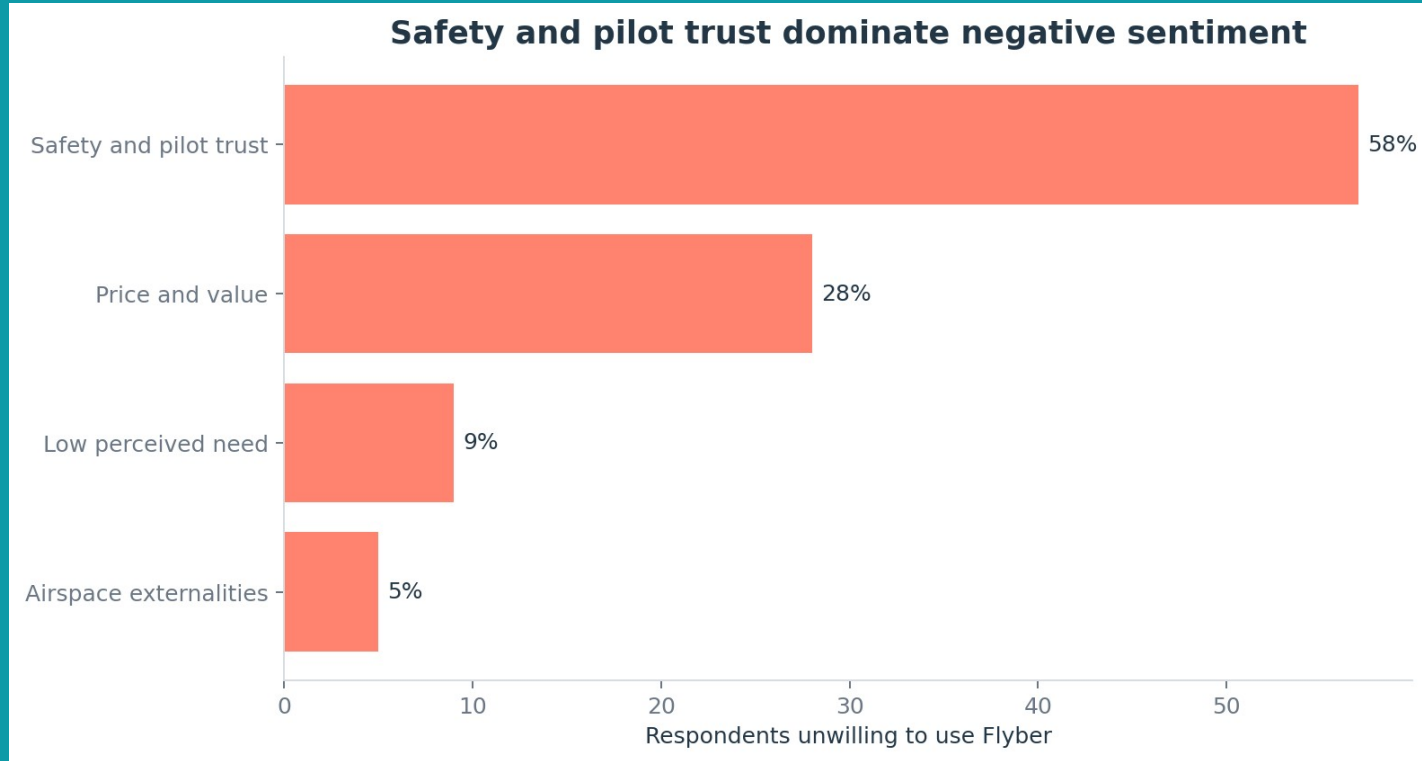
Taxi: 79.8% vs 81.7%; rideshare: 80.0% vs 80.4%. Neighborhoods: n >= 10.

Income separates WTP distributions



IQRs and medians; overall \$23/mile; neighborhood n >= 10.

Safety drives 57.6% of objections



Negative-response segmentation: n = 99; safety and pilot trust lead.

Hooray! End of Section 1.

You will complete Section 2 at the end of this course.

Please submit this file for review for Section 1.

Section 2: Proposal Synthesis

Identify a product objective for Flyber's launch. Your product objective will guide your KPIs, so identify what Flyber should optimize for. Your objective should be centered around one the following focus areas:

- User Acquisition
- User Engagement
- User Retention
- Profitability

Explain your reasoning. Include both why you feel your focus area is more relevant than the others for Flyber at this time of the product development cycle.

Objective: acquire qualified riders

Optimize USER ACQUISITION for time-sensitive one- and two-person trips in the four-node Manhattan/airport catchment.

WHY NOW: Flyber has no installed base. Test whether 80.2% stated interest becomes completed paid rides at feasible nodes, times, and prices.

ENGAGEMENT is premature before a repeatable first-use funnel; measure it diagnostically.

RETENTION requires an activated cohort; track 30-day repeat as a secondary metric.

PROFITABILITY follows demand and safety validation; measure cost now, but defer scale optimization.

Formulate 3-5 Key Performance Indicators (KPIs), to measure if the product is heading towards the right direction based on your objective

Four KPIs connect acquisition to trust

1. FIRST-RIDE CONVERSION = $\text{unique riders with trip_completed(first_ride=true)} / \text{unique eligible riders offered service}$.
2. ACTIVATED RIDERS = count distinct rider_id with a first trip_completed event, reported per week and cumulatively.
3. FIXED-PRICE BOOKING RATE = $\text{booking_confirmed} / \text{quote_shown}$ where $\text{quoted_price} = \$20/\text{mile}$ and minimum = \$60.
4. COMPLETION RATE = $\text{trip_completed} / \text{booking_confirmed}$; serious incidents = count distinct incident_id at severity $\geq$ serious.

Diagnostics: CAC = $\text{acquisition spend} / \text{activated riders}$; 30-day repeat = $\text{activated riders with another trip_completed within 30 days} / \text{eligible activated riders}$.

Create hypotheses around what thresholds your KPIs would need to hit in order to determine success

Growth must pass a safety gate

H1: treatment first-ride conversion $\geq 25\%$ versus a 20% planning baseline.

H2: enroll 1,340 qualified users, 670 per arm. At the planning rates, expect about 302 first rides: $(670 \times 20\%) + (670 \times 25\%) = 301.5$.

H3: fixed-price booking rate $\geq 50\%$. The \$20/mile offer is below the \$23/mile median among willing respondents.

GATE: completion $\geq 95\%$ and zero serious incidents. Any serious incident stops the experiment.

LEARN: report CAC and 30-day repeat, but do not claim unsupported pre-launch baselines.

As the product manager, you make decisions based on the insights you extract, we'll need to know the feature set we'll include in the MVP to measure viability, while keeping operational expenditure under control:

- What times/days of operation should the service run for?
- How many pick-up / drop-off nodes should we have?
- Where should the nodes be located?
- Should we initially use copters or homegrown hardware?
- Should the pricing be fixed or dynamic? At what rates?

A four-node, fixed-price MVP limits risk

WHEN: Thursday-Saturday, 5-11 PM; demand peaks at 6-7 PM and stays high through 10 PM.

NODES: East 34th Street, West 30th Street, Downtown/Pier 6, and LaGuardia connect key Manhattan demand to airport access.

AIRCRAFT: use contracted, certified copters and operators; defer homegrown hardware so the MVP tests demand rather than aircraft development.

PRICING: fixed \$20/mile with a \$60 minimum and no surge; defer dynamic pricing. The rate is below the \$23/mile median among willing respondents.

GATES: permits, weather, airspace, noise, accessibility, insurance, community acceptance, and unit economics.

Determine the MVP sample size & time period allotted estimated to come to a conclusion on your hypotheses.

1,340 users require a 32-week test

Optimizely

Products Solutions Resources Plans Developers Partners LoginGet started

A/B test sample size calculator

Powered by Optimizely Experimentation's stats engine.

Baseline conversion rate
Your control group's expected conversion rate [?]

20 %

Minimum Detectable Effect
The minimum relative change in conversion rate you would like to be able to detect.. [?]

25 %

Statistical Significance
95% is an accepted standard for statistical significance, although Optimizely allows you to set your own threshold for significance based on your risk tolerance.. [?]

95%

80%99%

Sample size per variation

670

Flyber Experiment Duration Calculator

The total sample exactly matches Optimizely: 670 users per variation x 2 variations = 1,340 users.

Total sample (users)	1,340	
Estimated qualifying rides per operating day	14	
Operating days per week	3	
Operating days required	95.7	
Calendar weeks required	31.9	
Planned duration (weeks)	32	

Formula: $\text{ceil}[(1,340 \text{ users} / 14 \text{ rides per operating day}) / 3 \text{ operating days per week}] = 32 \text{ weeks}$

Stop only after both arms reach 670 qualified users and all safety gates remain satisfied.

Create an instrumentation plan for the events you need collected and logged, in order to be able to physically measure your KPIs.

Each event is explicitly linked to a KPI

	A	B	C	D
1	Flyber KPI Instrumentation Plan			
2	Event	Exact trigger	Required properties	KPI and formula role
3	eligibility_viewed	Fires once when the app renders the launch offer after service-area, schedule, age, and consent rules pass.	event_id, event_time_utc, user_id, session_id, cohort_id, experiment_arm, node_from, node_to, eligibility_rule_version	First-Ride Conversion denominator; arm exposure count.
4	quote_shown	Fires when the rider sees a complete fixed-price quote before choosing whether to book.	event_id, shown_at, quote_id, user_id, session_id, node_from, node_to, quoted_price_usd, price_per_mile, minimum_fare_usd, experiment_arm	Fixed-Price Booking Rate denominator.
5	booking_confirmed	Fires after payment authorization succeeds and a reservation ID is created.	event_id, confirmed_at, booking_id, quote_id, user_id, trip_id, amount_usd, acquisition_source, experiment_arm	Fixed-Price Booking Rate numerator; Completion Rate denominator.
6	riderPickedUp	Triggers whenever the driver taps the 'Passenger is picked up' button on the Flyber driver app.	passenger_count, rider_id, driver_id, pickup_timestamp, pickup_latitude, pickup_longitude	Completion Rate validation; supplies the pickup-to-completion diagnostic denominator.
7	trip_completed	Fires when the aircraft arrives and the operator closes the trip with status = completed.	event_id, completion_time, trip_id, booking_id, rider_id, node_from, node_to, aircraft_id, operator_id, first_ride_flag, ride_sequence	First-Ride Conversion and Activated Riders numerator; Completion Rate numerator; 30-day repeat numerator.
8	incident_logged	Fires when an operator or safety officer opens the incident workflow and saves the initial record.	event_id, reported_at, incident_id, trip_id, reporter_role, category, severity, review_status	Serious-Incident safety gate; any severity = serious event stops the experiment.
9				
10	KPI formulas from events			
11	First-Ride Conversion = $\text{DISTINCT}(\text{first-ride trip_completed.rider_id}) / \text{DISTINCT}(\text{eligibility_viewed.user_id})$			
12	Activated Riders/week = $\text{DISTINCT}(\text{first-ride trip_completed.rider_id})$, grouped by operating week			
13	Fixed-Price Booking Rate = $\text{COUNT}(\text{booking_confirmed}) / \text{COUNT}(\text{quote_shown})$			
14	Completion Rate = $\text{COUNT}(\text{trip_completed}) / \text{COUNT}(\text{booking_confirmed})$; pickup audit = $\text{COUNT}(\text{trip_completed}) / \text{COUNT}(\text{riderPickedUp})$; serious incidents = $\text{DISTINCT incident_logged.incident_id WHERE severity = serious}$			

Create a qualitative feedback survey questions for users after their ride, to further understand and optimize the product for future iterations.

Post-ride feedback explains the metrics

MECHANIC: send within 15 minutes of arrival; mobile form under three minutes; optional interview follow-up; link by survey_id to trip and cohort, not free-text identity.

1. How likely are you to ride Flyber again in the next 30 days? 0 (not at all) to 10 (extremely likely). Why?
2. Did anything go wrong? Select all: safety concern, delay, cancellation, node access, booking/payment, comfort, price, accessibility, other. Describe the most important issue.
3. Rate safety/trust, node access, travel time, price/value, and support from 1 (poor) to 5 (excellent).
4. Which should Flyber improve or launch next? Rank: more nodes, longer hours, lower/flexible price, booking changes, comfort, accessibility, disruption support.
5. What single change would most increase your likelihood of riding again?

Summarize everything you have learned into your final proposal

- Identify the target population. Why did you select that target population? What are their pain points?
- Create a product proposal containing claim, evidence, estimated impact, and risks
- Claims should be backed by quantitative evidence, impact should assess market needs/benefits
- Risks involve any known unknowns that we'll still need to monitor post-launch
- State cross-functional stakeholder teams that will need to be involved

Target slow, time-sensitive short trips

TARGET: adults making one- or two-person, time-sensitive trips near Midtown, Lower Manhattan, and LaGuardia during Thursday-Saturday evenings.

WHY: 85.3% of valid taxi rides carry 1-2 passengers; Midtown NTAs lead endpoint density and minutes-per-mile; 80.2% of valid survey responses state interest.

PAIN: unpredictable road delay, missed connections, surge/cancellation uncertainty, and poor time reliability on short congested routes.

SEGMENTATION: prioritize observed trip context and proximity to nodes, not demographic stereotypes. The survey is unweighted and measures stated preference.

Proposal: test four nodes for 32 weeks

CLAIM: four-node evening service at fixed \$20/mile can convert high-friction Manhattan/airport trips into paid first rides.

EVIDENCE: 85.3% one- or two-rider share; Midtown density/delay; 80.2% interest; \$23/mile median willingness to pay.

IMPACT: enroll 1,340 qualified users, 670 per arm; expect about 302 first rides at the planning rates; run up to 32 weeks at 14 qualifying rides per operating day.

KNOWN UNKNOWNNS: conversion, capacity, repeat demand, unit margin, permits, weather, accessibility, noise response, and serious-event risk.

RISK: failure could waste capital or harm safety and trust; success establishes bounded demand evidence. Scale only if all gates pass.

Cross-functional ownership is explicit

PRODUCT + RESEARCH: objective, experiment design, eligibility, post-ride learning, and roadmap.

DATA + ENGINEERING: schemas, instrumentation, identity, dashboards, QA, privacy, booking, payments, and reliability.

FLIGHT OPS + SAFETY: operator qualification, dispatch, weather, aircraft readiness, incident command, and stop-work authority.

LEGAL + REGULATORY + SECURITY: permits, insurance, accessibility, privacy, cyber risk, and agency coordination.

COMMERCIAL + COMMUNITY: unit economics, contracts, node operators, acquisition, support, service recovery, noise engagement, and public trust.

Interactive Tableau workbook

Explore the full enriched taxi dataset through the published pickup-opportunity map.

TABLEAU PUBLIC VIEW:

[https://public.tableau.com/app/profile/christopher.ohara/viz/
FlyberMVPLaunchStrategy_17861346231600/PickupOpportunityMap#1](https://public.tableau.com/app/profile/christopher.ohara/viz/FlyberMVPLaunchStrategy_17861346231600/PickupOpportunityMap#1)

Select map marks or neighborhood legend entries to inspect the NYC pickup geography interactively.